

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

اللَّهُمَّ إِنِّي أَسْأَلُكَ عِلْمًا نَافِعًا،

وَرِزْقًا طَيِّبًا، وَعَمَلًا مُتَقَبَّلًا،

Course Plan MATH2501



Introduction to Statistics

Measures of Central Tendency

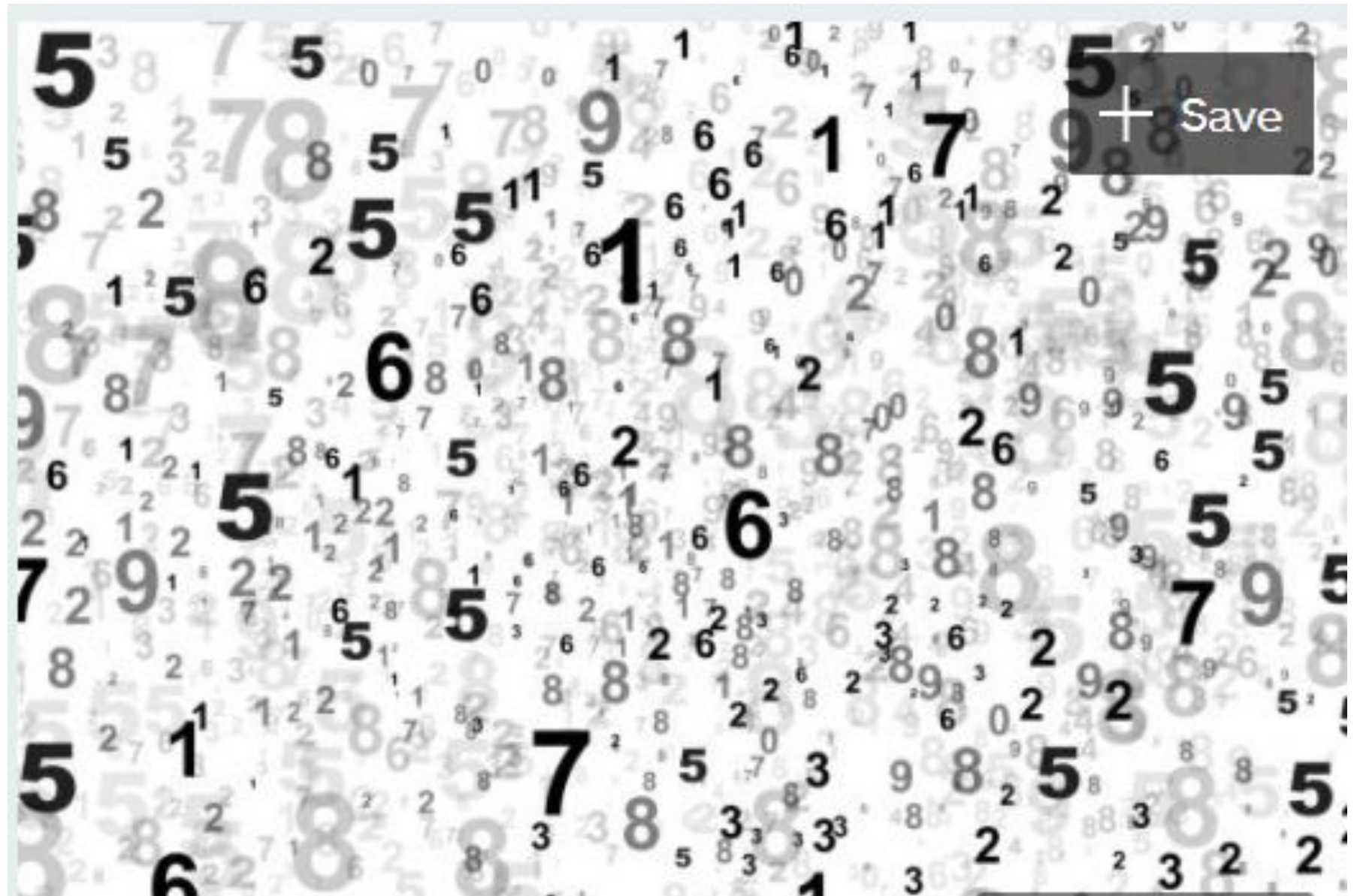
Week 1

Statistics



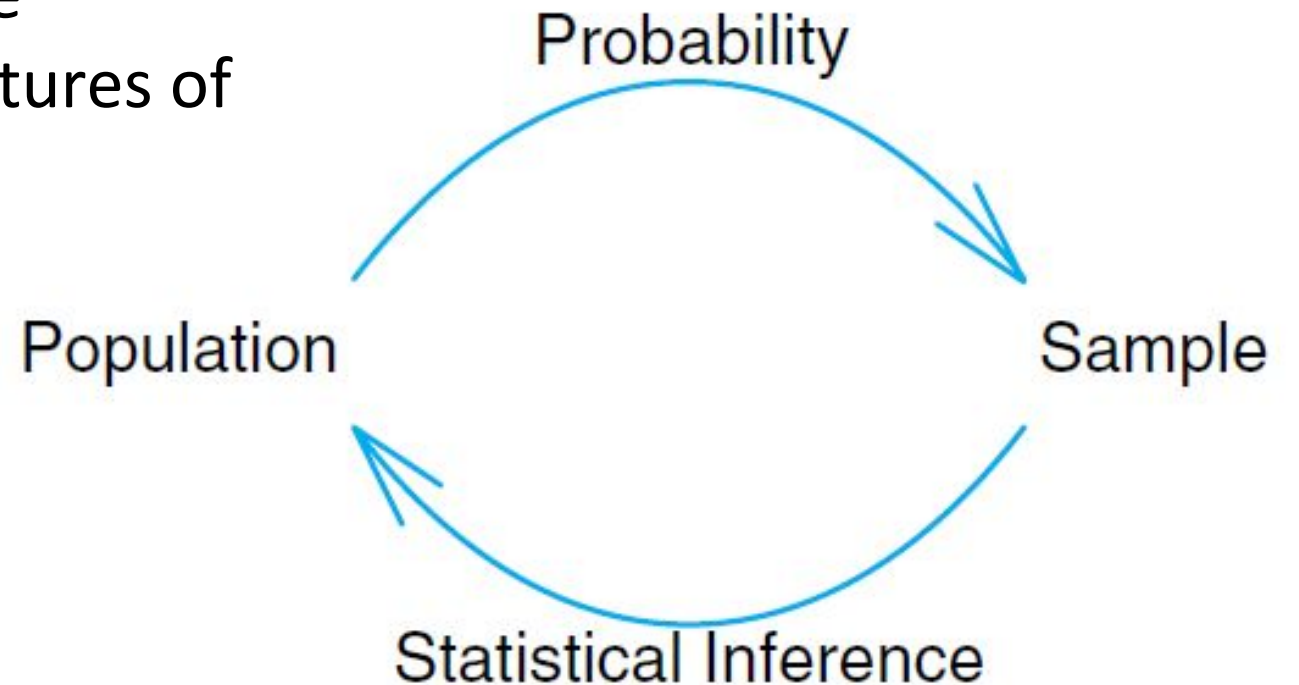
<https://www.youtube.com/watch?v=MXaJ7sa7q-8>

DATA



Probability and Statistical Inference

Elements in probability allow us to draw conclusions about characteristics of hypothetical data taken from the population, based on known features of the population.



Collection of data

- Simple random sampling
- Stratified
- Cluster
- Systematic
- Circular
- ...



Learning Intentions

Today we will understand:

- ▶ Measures of Central Tendency
 - * Mean
 - * Median
 - * Mode



Measures of Location

<https://study.com/academy/lesson/central-tendency-measures-definition-examples.html>

► Arithmetic average

$$\text{Mean} = \frac{\text{Sum of all data values}}{\text{Number of data values}}$$

Symbolically,

$$\bar{x} = \frac{\sum x}{n}$$

where:

$\bar{x}$ (read as 'x bar') is the mean of the set of x values

$\sum x$ is the sum of all the x values

n is the number of x values



Sample Mean

Suppose that the observations in a sample are $x_1, x_2, \dots, x_n$. The **sample mean**, denoted by $\bar{x}$, is

$$\bar{x} = \sum_{i=1}^n \frac{x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}.$$

- ▶ Calculate the mean height of JCU students

$$\bar{x} = \frac{\sum x}{n}$$

$$= \frac{(175 + 163 + 155 + 149 + 180 + 165)}{6}$$

$$= \frac{987}{6}$$

$$= 164.5 \text{ cm}$$

Height (cm)
175
163
155
149
180
165

1)

Weight (g)
23
50
16
44
36
29
47
52
35
42
41
28
26
24
38

2)

Time (s)
90
63
87
56
71
38
42
52
86
79
67
83
69
82
64

3)

Length (m)
5.3
2.8
3.6
3.9
2.7
4.8
4.2
6.2
5.9
5.1
4.6
5.3
6.9

$$1) \quad \bar{x} = \frac{\sum x}{n}$$

$$= \frac{(23 + 50 + 16 + 44 + 36 + 29 + 47 + 52 + 35 + 42 + 41 + 28 + 26 + 24 + 38)}{15}$$

$$= 531$$

$$= 34.5 \text{ g}$$

$$2) \quad \bar{x} = \frac{\sum x}{n}$$

$$= \frac{(90 + 63 + 87 + 56 + 71 + 38 + 42 + 52 + 86 + 79 + 67 + 83 + 69 + 82 + 64)}{15}$$

$$= 1029$$

$$= 68.6 \text{ s}$$

$$3) \quad \bar{x} = \frac{\sum x}{n}$$

$$= \frac{(5.3 + 2.8 + 3.6 + 3.9 + 2.7 + 4.8 + 4.2 + 6.2 + 5.9 + 5.1 + 4.6 + 5.3 + 6.9)}{13}$$

$$= 613$$

$$= 4.7 \text{ m}$$

Median

Given that the observations in a sample are $x_1, x_2, \dots, x_n$, arranged in **increasing order** of magnitude, the sample median is

$$\tilde{x} = \begin{cases} x_{(n+1)/2}, & \text{if } n \text{ is odd,} \\ \frac{1}{2}(x_{n/2} + x_{n/2+1}), & \text{if } n \text{ is even.} \end{cases}$$

- ▶ If there is an odd number of values in the data set, then the median is the middle value

For the data set:

6, 9, 1, 2, 6, 5, 1

Arrange from lowest to highest:

1, 1, 2, 5, 6, 6, 9



The middle value

The median is 5

- ▶ If there is an even number of values in the data set, then the median is the mean of the two middle values

For the data set:

6, 9, 1, 2, 6, 5, 1

Arrange from lowest to highest:

1, 1, 2, 6, 6, 9

Two middle values

The median is the mean of 2 and 6:

$$\frac{2 + 6}{2} = 4$$

The median
is 4

► Determine the median for the following data sets

1) 132, 139, 131, 138, 132, 139, 133, 137, 139

2) 25, 10, 16, 25, 12, 22, 20, 23, 13, 10

3) 56, 23, 48, 78, 94, 35, 88, 69, 44, 53, 27

1) 132, 139, 131, 138, 132, 139, 133, 137, 139

Rearrange from lowest to highest:

131, 132, 132, 133, 137, 138, 139, 139, 139

Middle value - 137

Median = 137

2) 25, 10, 16, 25, 12, 22, 20, 23, 13, 10

Rearrange from lowest to highest:

10, 10, 12, 13, 16, 20, 22, 23, 25, 25

Middle values - 16 and 20

$$\text{Median} = \frac{16 + 20}{2} = 18$$

Median = 18

3) 56, 23, 48, 78, 94, 35, 88, 69, 44, 53, 27

Rearrange from lowest to highest:

23, 27, 35, 44, 48, 53, 56, 69, 78, 88, 94

Middle value - 53

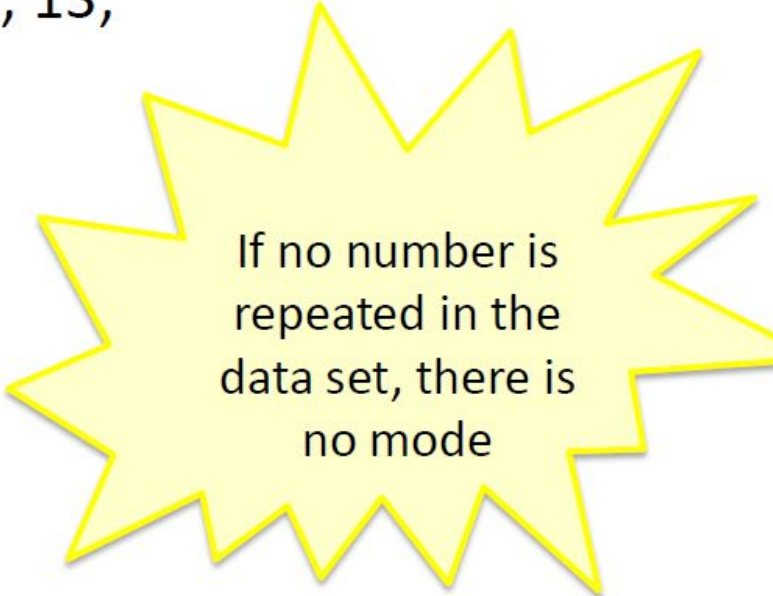
Median = 53

Mode

- ▶ The most frequent measurement

12, 11, 15, 12, 12, 11, 14, 17, 15, 12, 13,

Number of Cars Sold	Frequency
11	2
12	4
13	1
14	1
15	2
17	1



If no number is repeated in the data set, there is no mode

► Determine the mode for the following data sets

1) 132, 139, 131, 138, 132, 139, 133, 137, 139

2) 3, 3, 3, 5, 5, 5, 3, 6, 4, 8, 5, 4, 2, 4, 3, 5

3) 56, 23, 48, 78, 94, 35, 88, 69, 44, 53, 27

1) 132, 139, 131, 138, 132, 139, 133, 137, 139

= 139

2) 3, 3, 3, 5, 5, 5, 3, 6, 4, 8, 5, 4, 2, 4, 3, 5

= 3 and 5 = bimodal

3) 56, 23, 48, 78, 94, 35, 88, 69, 44, 53, 27

= no mode

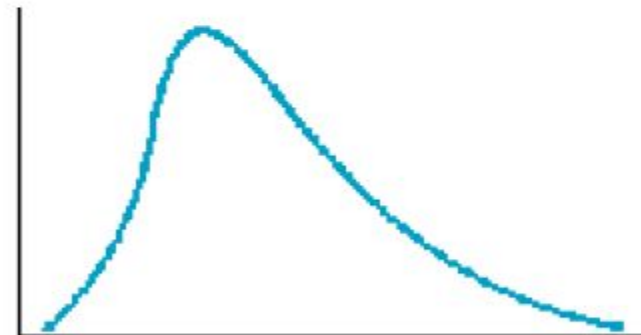
- ▶ Your data will determine which measure of central tendency is appropriate
- ▶ PLOT DATA FIRST



**Negatively (left)
skewed
distribution**

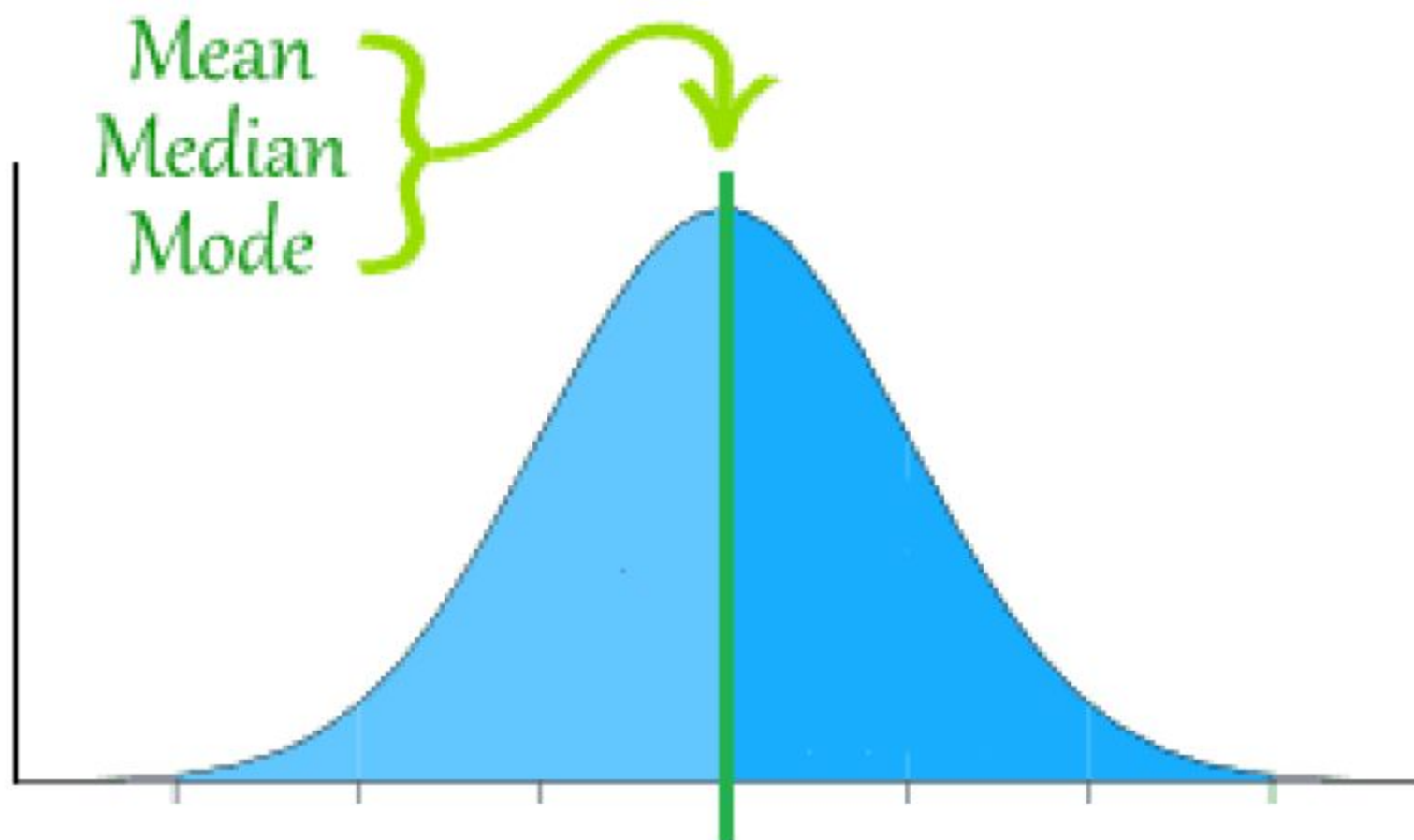


**Normal
skewed
distribution**



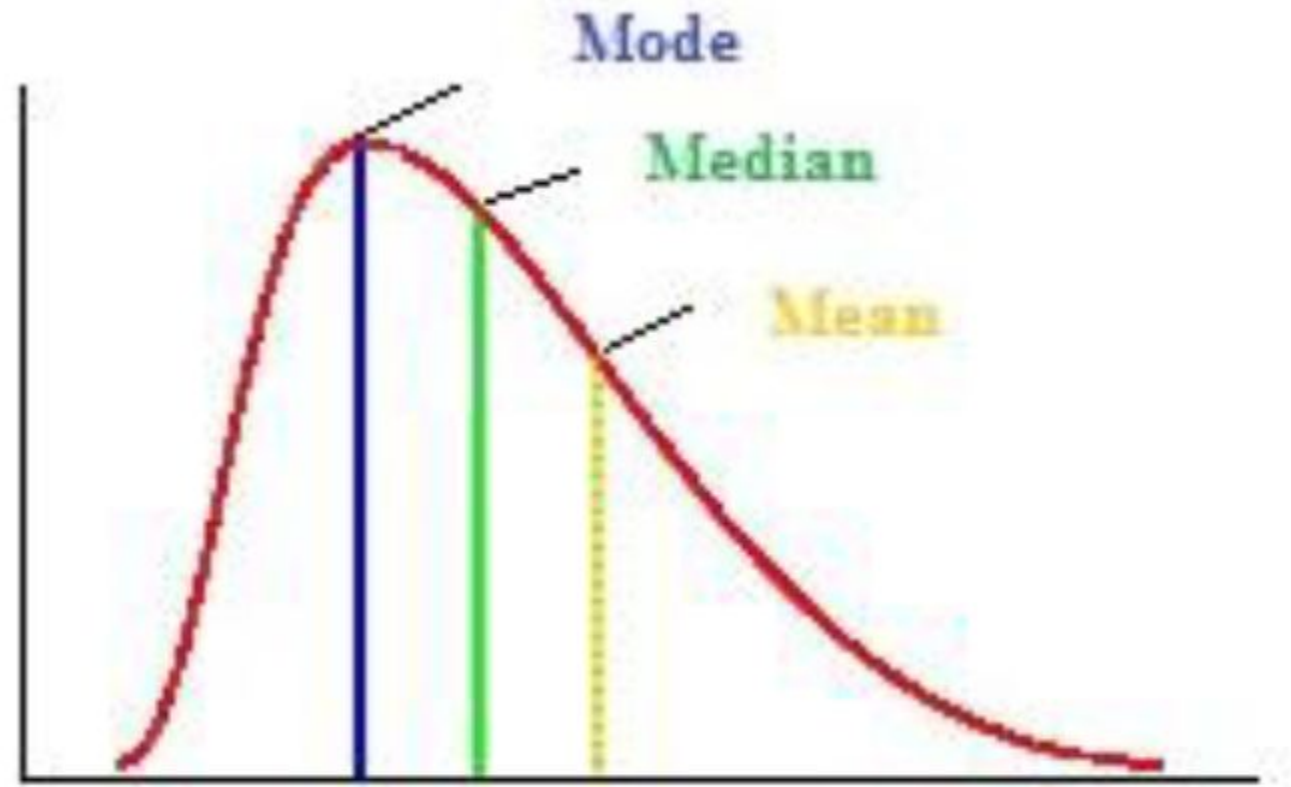
**Positively (right)
skewed
distribution**

- ▶ Normally distributed data



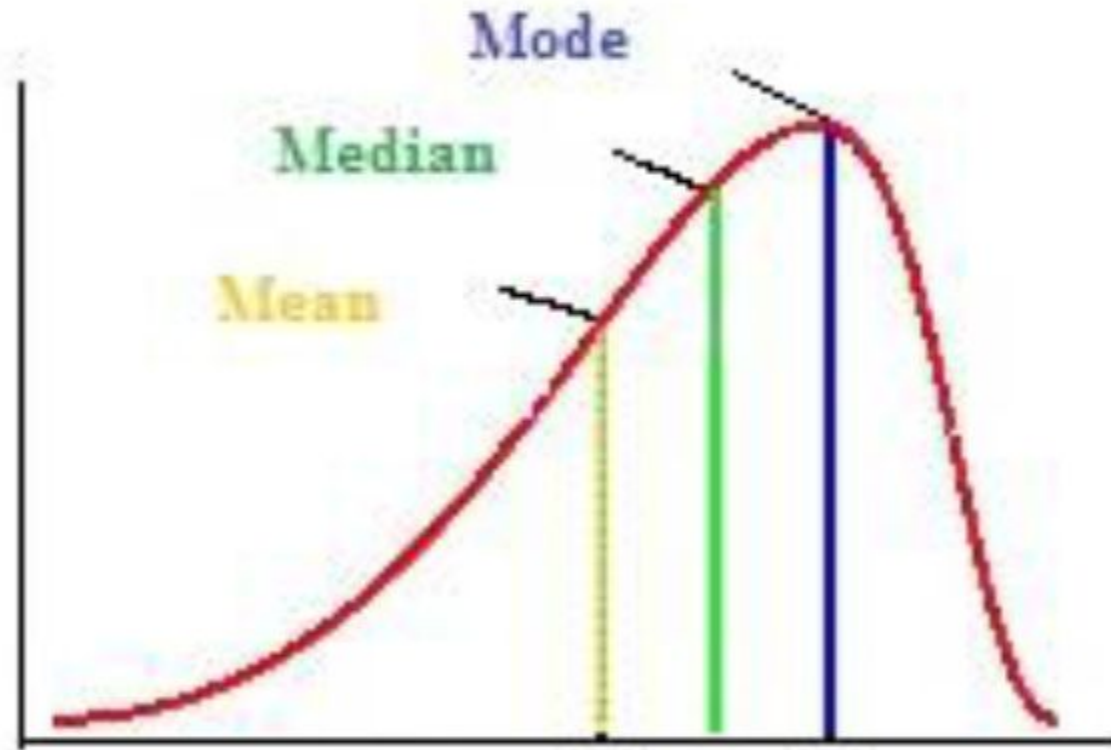
Normally Distributed Data: MEAN = MEDIAN = MODE

- ▶ Positively skewed (right) distribution



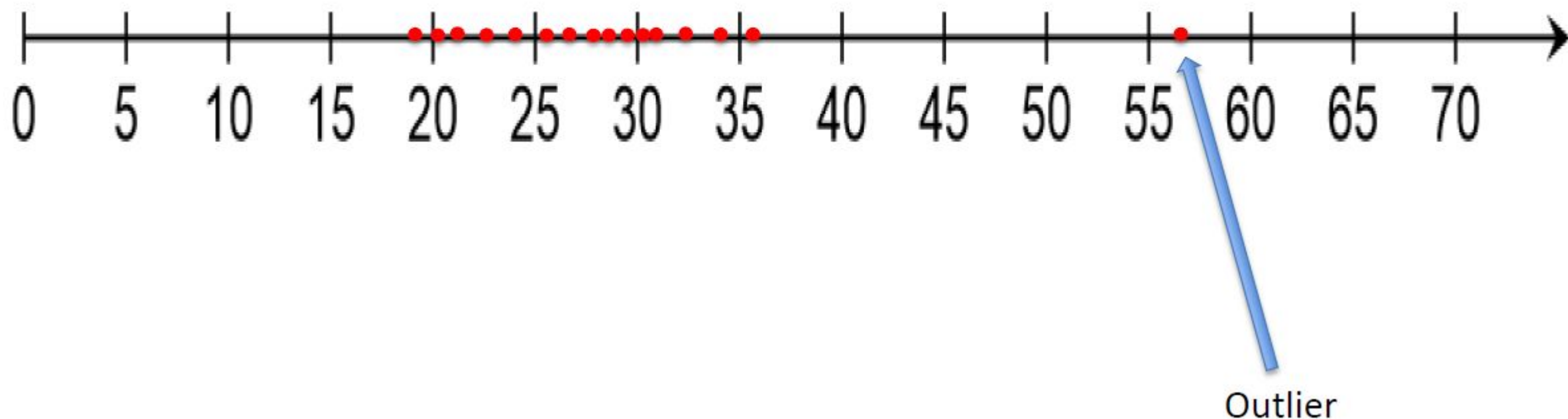
Skewed Right: $MODE < MEDIAN < MEAN$

- ▶ Negatively skewed (left) distribution



Skewed Left: MEAN < MEDIAN < MODE

- ▶ Data values that are 'far away' from the main group of data
- ▶ **Outliers** are the values that **lie outside** the other values



- ▶ Outliers have **extreme effects** on the **mean**

Consider:

5, 6, 4, 7, 6, 19

$$\bar{x} = \frac{\sum x}{n}$$

$$= \frac{5 + 6 + 4 + 7 + 6 + 19}{6}$$

Mean = 9

5, 6, 4, 7, 6

$$\bar{x} = \frac{\sum x}{n}$$

$$= \frac{5 + 6 + 4 + 7 + 6}{5}$$

Mean = 5.6

- ▶ Outliers **do not** have **extreme effects** on the **median**

Consider:

5, 6, 4, 7, 6, 19

5, 6, 4, 7, 6

Arrange from lowest to highest:

4, 5, 6, 6, 7, 19

4, 5, 6, 6, 7

= 6

= 6

Example

The following measurements were recorded for the drying time, in hours, of a certain brand of latex paint.

3.4 2.5 4.8 2.9 3.6 2.8 3.3 5.6 3.7 2.8 4.4 4.0 5.2 3.0 4.8

Assume that the measurements are a simple random sample.

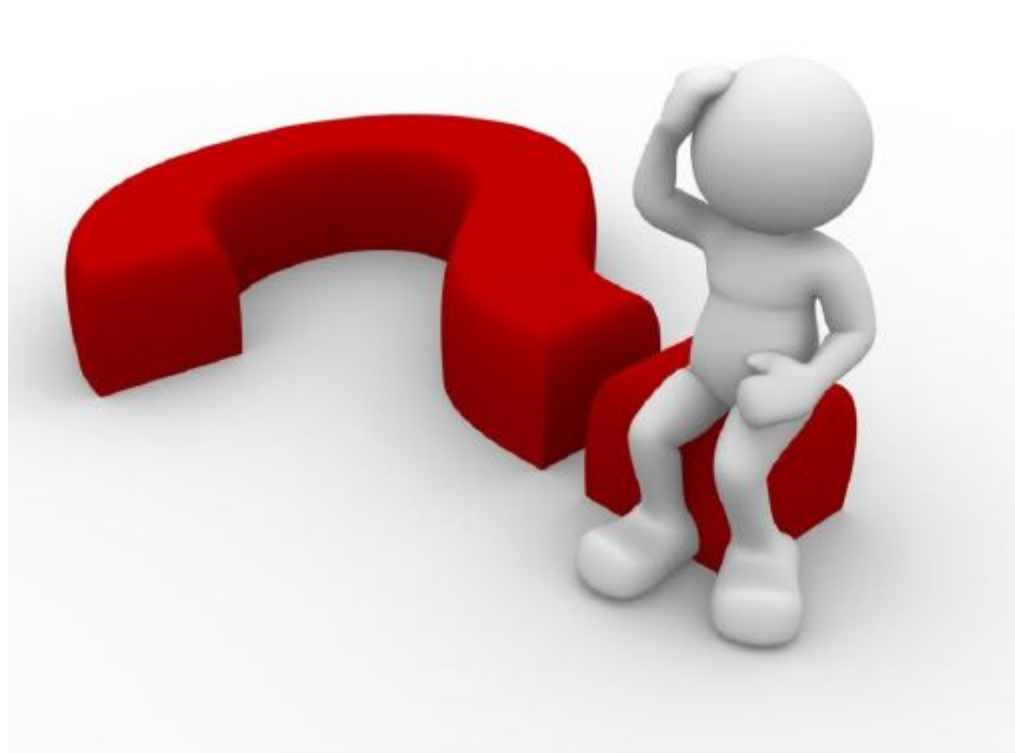
- (a) What is the sample size for the above sample?
- (b) Calculate the sample mean for these data.
- (c) Calculate the sample median.
- (e) Compute the 20% trimmed mean for the above data set.
- (f) Is the sample mean for these data more or less descriptive as a center of location than the trimmed mean?

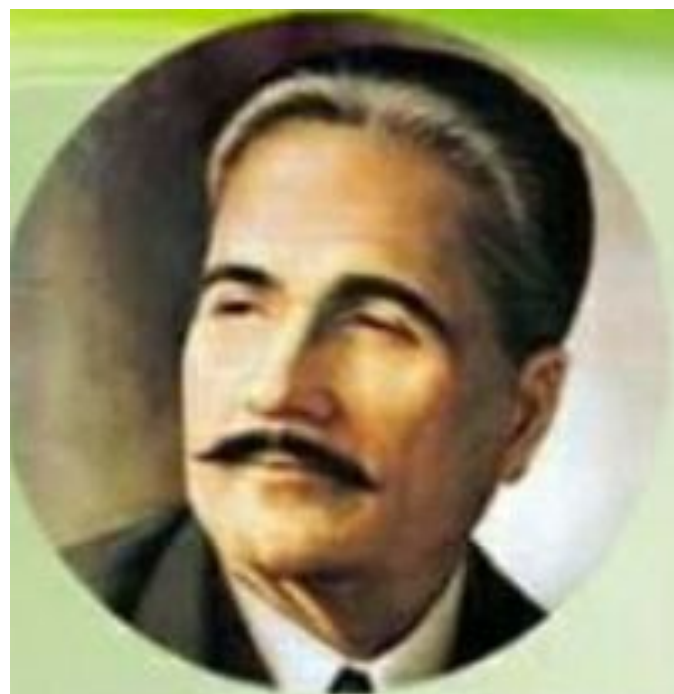
TASK

1.2 According to the journal *Chemical Engineering*, an important property of a fiber is its water absorbency. A random sample of 20 pieces of cotton fiber was taken and the absorbency on each piece was measured. The following are the absorbency values:

18.71	21.41	20.72	21.81	19.29	22.43	20.17
23.71	19.44	20.50	18.92	20.33	23.00	22.85
19.25	21.77	22.11	19.77	18.04	21.12	

- (a) Calculate the sample mean and median for the above sample values.
- (b) Compute the 10% trimmed mean.
- (c) Do a dot plot of the absorbency data.
- (d) Using only the values of the mean, median, and trimmed mean, do you have evidence of outliers in the data?





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(اقبال)